
Homework 3

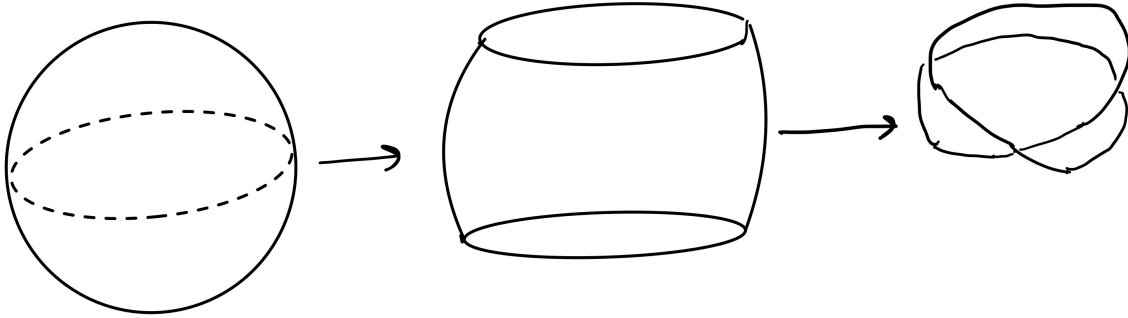
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Problem 1

Since $\mathbb{RP}^2$ identifies antipodal points with each other, removing a disc from $\mathbb{RP}^2$ is the same as removing two discs from the sphere, and identifying antipodal points on the resultant figure.



Problem 2

Note that if $[(z, w)] \in \mathbb{CP}^1$ with $z \neq 0$, then $[(z, w)] = [(1, w/z)]$. Thus, every such point with $z \neq 0$ can be represented of this form. If $z = 0$, then we instead have $[(0, w)] = [(0, w/w)] = [(0, 1)]$ since $(z, w) \neq (0, 0)$. Any two points $[(1, w/z)], [(1, a/b)]$ are now distinct in $\mathbb{CP}^1$ if $w/z \neq a/b$, since multiplying through by a scaling factor would also shift the value of the 1. We can thus view $\mathbb{CP}^1$ as $\mathbb{C} \cup \{(0, 1)\}$.

Let $\phi : \mathbb{CP}^1 \rightarrow S^2$ be the map $\phi(0, 1) = (1, 0, 0)$, and

$$\phi(1, z) = \left(\frac{2 \operatorname{Re} z}{|z|^2 + 1}, \frac{2 \operatorname{Im} z}{|z|^2 + 1}, \frac{|z|^2 - 1}{|z|^2 + 1} \right).$$

This map is well-defined by the logic I explained previously, identifying each element of $\mathbb{CP}^1$ with an element of the form $[(1, z)]$ or $[(0, 1)]$. Let $\psi : S^2 \rightarrow \mathbb{CP}^1$ be the map

$$\psi(x, y, z) = \left[\left(1, \frac{x + iy}{1 - z} \right) \right],$$

and $\psi(0, 0, 1) = [(0, 1)]$. These maps are both obviously well-defined, and a lot of explicit computation shows they are in fact inverses. Therefore, they are bijections.

Since ϕ is a composition of continuous functions (rationals, Re , Im , etc.), it is continuous on elements other than $(0, 0, 1)$.

ψ is seen to be continuous at points other than $(0, 0, 1)$ for the same reasons. As a point approaches $(0, 0, 1)$ on S^2 , we have x, y approaching 0, and thus $\psi(x, y, z)$ approaches $[(0, 1)]$. Therefore, ψ is continuous.

Problem 3

Let $A \subseteq X$, $B \subseteq Y$ be closed sets. Then $A = \overline{A}$, $B = \overline{B}$. We want to show $A \times B = \overline{A \times B}$. Let $(x, y) \in \overline{A \times B}$. Then every open set U containing (x, y) intersects $A \times B$ at some point. Note that the projection $\pi : X \times Y \rightarrow X$ given by $(x, y) \mapsto x$ is continuous, as I showed in the previous homework. Let $x \in V \subseteq X$ be open. Note that $(x, y) \mapsto x$, so $(x, y) \in \pi^{-1}(V)$. Since every open set containing (x, y) intersects $A \times B$, it follows that there exists $(x', y') \in \pi^{-1}(V) \cap (A \times B)$. We thus have $\pi(x', y') = x' \in V \cap A$, and thus every open set containing x intersects A at some point x' . Therefore, $x \in \overline{A} = A$. By equivalent logic, $y \in B$.

Thus, for all $(x, y) \in \overline{A \times B}$, we have $x \in A$ and $y \in B$, and thus $(x, y) \in A \times B$. Thus, $\overline{A \times B} \subseteq A \times B$. We also have $A \times B \subseteq \overline{A \times B}$ by definition, and thus $A \times B = \overline{A \times B}$. Thus, $A \times B$ is closed.

Problem 4

- If $d(\{x\}, A) = 0$, then $\inf \{d(x, a) \mid a \in A\} = 0$. Therefore, for all $a \in A$, $d(x, a) \geq 0$. Suppose for contradiction that $d(x, a) > 0$ for all $a \in A$. If there were to exist some $a \in A$ such that $d(x, a)$ were a lower bound for this, meaning $d(x, a) \leq d(x, b)$ for all $b \in A$, then we would have $d(x, a) = \inf \{d(x, a) \mid a \in A\}$ by definition, and thus $d(x, a) = 0$, a contradiction. Therefore, there exists no lower bound $d(x, a)$ with $a \in A$, and we must have that for any $a \in A$, there exists some $a' \in A$ such that $d(x, a') < d(x, a)$. We must also have that for any $r < 0$, there exists $a \in A$ such that $d(x, a) < r$ by definition of infimum. Therefore, for all $r \in \mathbb{R}^+$, there exists some $B_s(x)$ with $s = d(x, a) < r$ for some $a \in A$, such that $x \in A$.

Since every open set containing x must be a union of basis sets, by logic used in the previous homework we can reduce the question of all open sets containing x to that of all basis sets containing x . Let $B_\ell(z)$ be an open ball containing x , and note that $B_{\ell-d(x,z)}(x) \subseteq B_\ell(z)$ by logic used extensively in the previous homework that I don't want to type back up. Since $\ell - d(x, z) \in \mathbb{R}^+$, there exists some $a \in A$ such that $d(x, a) < \ell - d(x, z)$, and thus $a \in B_{\ell-d(x,z)}(x)$. Therefore, $B_\ell(z)$ contains a . Therefore, every open set containing x intersects A at some point, and $x \in \overline{A}$. Since A is closed, $x \in A$. This contradicts our assumption that $d(x, a) > 0$ for all $a \in A$, and thus there exists $a \in A$ such that $d(x, a) = 0$. By definition of a metric, we have $x = a$, and thus $x \in A$.

- Let $(a, b) = B_r(x)$ be an open interval in $\mathbb{R}$, and note that $a \notin (a, b)$. Suppose for contradiction that $\inf \{d(a, y) \mid y \in B_r(x)\} > 0$. It follows that there exists $y \in B_r(x)$ such that $d(a, y) > 0$, and $d(a, z) \geq d(a, y)$ for all $z \in B_r(x)$. Since $a < x$, we must have $z < x$, otherwise $d(x, a) < d(z, a)$, which would be a contradiction. Define

$$z := x - \left(r - \frac{d(a, y)}{2}\right).$$

Note that

$$d(a, x) = \left|x - x + r - \frac{d(a, y)}{2}\right| = r - \left|\frac{d(a, y)}{2}\right| < r.$$

Therefore, $z \in B_r(x)$. Moreover, note that

$$d(a, z) = \left|(x - r) - x + r - \frac{d(a, y)}{2}\right| = \frac{d(a, y)}{2} < d(a, y).$$

Therefore, $d(a, z) < d(a, y)$, contradicting our assumption that $\inf \{d(a, y) \mid y \in B_r(x)\} > 0$.

- Let $x, y \in X$ such that $x \neq y$. Let $d(x, y) = r$. Then $y \notin B_{r/2}(x)$, since $d(x, y) = r > r/2$, and similarly $x \notin B_{r/2}(y)$. Now let $a \in B_{r/2}(x)$. We must have $d(x, a) < r/2$, and thus

$$d(x, y) \leq d(x, a) + d(a, y) \implies d(a, y) \geq d(x, y) + d(x, a) = r + d(x, a) > r.$$

Since $d(y, a) > r > r/2$, we have that $a \notin B_{r/2}(y)$. By the same logic, for any $a \in B_{r/2}(y)$, we have $a \notin B_{r/2}(x)$. Therefore, $B_{r/2}(x) \cap B_{r/2}(y) = \emptyset$, and thus X is Hausdorff.

By the first homework problem,

$$\bigcap_V \bar{V} = \{x\},$$

where V ranges over open sets containing x . Since this is an intersection of closed sets, we have that $\{x\}$ is closed for all $x \in X$.

- Let $Y \subseteq X$ be closed, and let $x \in X$ such that $x \notin Y$. By contrapositive of part 1, since $x \notin Y$, then either Y is not closed, or $d(\{x\}, Y) \neq 0$. Since the first statement is false, we must have $d(\{x\}, Y) > 0$. Define $d(\{x\}, Y)/2 =: r$. Let $U := B_r(x)$, and let

$$V := \bigcup_{y \in Y} B_r(y).$$

We clearly have that $x \in U$ and $Y \subseteq V$, so it suffices to show these sets are disjoint.

Let $a \in U$, and let $y \in Y$. We will show $a \notin B_r(y)$. We have

$$d(x, y) \leq d(x, a) + d(a, y) \implies d(a, y) \geq d(x, y) - d(x, a).$$

Since $2r = \inf \{d(x, y) \mid y \in Y\}$, it follows that $d(x, y) \geq 2r$ for all $y \in Y$. Thus,

$$d(a, y) \geq d(x, y) - d(x, a) \geq 2r - d(x, a) > 2r - r = r.$$

Therefore, $d(a, y) > r$, and thus $a \notin B_r(y)$.

Now let $a \in V$. It follows that there exists some $B_r(y)$ in the union such that $a \in B_r(y)$. By the same logic, $d(x, a) \geq 2r$, and thus

$$d(x, y) \leq d(x, a) + d(a, y) \implies d(x, a) \geq d(x, y) - d(a, y) \leq 2r - d(a, y) > 2r - r = r.$$

Therefore, $d(x, a) > r$, and $a \notin B_r(x)$. Therefore, $U \cap V = \emptyset$, and X is T_3 .

Problem 5

For notational purposes, let $\mathcal{B}$ be the metric basis, and let $\mathcal{B}_Y$ be the set in question $\{B_r(y) \mid r \in \mathbb{Q}, y \in Y\}$. I also don't remember us showing the following result in class, so I will show it here.

Lemma 1. *Let $Y \subseteq X$ be a dense subset of some topological space X . Then for all nonempty open sets $U \subseteq X$, there exists $y \in U \cap Y$. That is, every nonempty open set contains some element of Y .*

Proof. Since Y is dense, by definition $\bar{Y} = X$. Since $\bar{Y} = X$ is the smallest closed set containing Y , it follows that no proper closed subset $U \subsetneq X$ contains Y . Equivalently, if $U \subsetneq X$ is closed, then $Y \setminus U \neq \emptyset$.

If $Y \cap U = \emptyset$, then $U^c \subseteq Y$, and since $U \neq X$, we have $U^c = U^c \cap Y \neq \emptyset$, and we are done. If U and Y are not disjoint, then $Y \setminus U \neq \emptyset$ since U doesn't contain Y . By definition, this implies $Y \cap U^c \neq \emptyset$. Since U^c is open, there exists $y \in Y \cap U^c$. Since every nonempty open set can be represented as U^c for $U \subsetneq X$ closed, we have that every nonempty open set $U \subseteq X$ has some $y \in Y \cap U$. ■

Let $a \in X$. Since the set of open balls forms a basis for the metric topology, there exists some $B_r(x) \in \mathcal{B}$ such that $a \in B_r(x)$. Since Y is dense, there exists $y \in Y$ such that $y \in B_r(x)$. It follows that

$$d(a, y) \leq d(a, x) + d(a, y) < d(a, x) + r.$$

Therefore,

$$a \in B_{d(a, x) + r}(y).$$

Thus, for all $x \in X$, there exist $y \in Y$ and $r \in \mathbb{R}$ such that $a \in B_r(y)$. Note that since $\mathbb{Q}$ is dense in $\mathbb{R}$, for all $r \in \mathbb{R}$ there exists some $q \in \mathbb{Q}$ such that $q > r$. We thus have $B_r(y) \subseteq B_q(y)$, and $B_q(y) \in \mathcal{B}_Y$.

Now let $r, s \in \mathbb{Q}$ and $y_1, y_2 \in Y$. Then $B_r(y_1), B_s(y_2) \in \mathcal{B}_Y$. Then since $\mathcal{B}_Y \subseteq \mathcal{B}$, these sets are open, and thus $B_r(y_1) \cap B_s(y_2)$ is open. Since Y is dense, there exists $y \in Y$ such that $y \in B_r(y_1) \cap B_s(y_2)$. Define $\ell := \min\{r - d(y, y_1), s - d(y, y_2)\}$. Note that for all $a \in B_\ell(y)$, we have $d(a, y) < \ell$, and thus

$$d(a, y) < r - d(y, y_1) \implies d(a, y) + d(y, y_1) < r \implies d(a, y_1) \geq d(a, y) + d(y, y_1) < r.$$

Therefore, $a \in B_r(y_1)$. Similarly,

$$d(a, y) < s - d(y, y_2) \implies d(a, y) + d(y, y_2) < s \implies d(a, y_2) \leq d(a, y) + d(y, y_2) < s.$$

Therefore, $a \in B_s(y_2)$. Therefore, we have $B_\ell(y) \subseteq B_r(y_1) \cap B_s(y_2)$.

If $\ell \in \mathbb{Q}$, then $B_\ell(y) \in \mathcal{B}_Y$, and we are done. If not, then since $d(y, a) < \ell$, the set $(d(y, a), \ell)$ is open in $\mathbb{R}$. Since $\mathbb{Q}$ is dense in $\mathbb{R}$, there exists $q \in (d(y, a), \ell)$ such that $q \in \mathbb{Q}$, and thus $a \in B_q(y)$. Moreover, since $q < \ell$, we have $B_q(y) \subseteq B_\ell(y)$. Since $q \in \mathbb{Q}$, we have $B_q(y) \in \mathcal{B}_Y$, and we are done.

Problem 6

It suffices to show the identity map $\text{id} : (B_X, d_H^{d_1}) \rightarrow (B_X, d_H^{d_2})$ is continuous, since the logic in both directions is the same, thus making id a homeomorphism, and thus id gives a bijection $\mathcal{T}_1 \rightarrow \mathcal{T}_2$. Since it is the identity map, we would then have $\mathcal{T}_1 = \mathcal{T}_2$.

Let $A \in \mathcal{B}_1$, and fix $\varepsilon > 0$. Note that for all $B \in B_\varepsilon^1(A)$, we must have that for all $b \in B$, there exists $a \in A$ such that $d_1(a, b) < \varepsilon$. Since the topologies induced by d_1 and d_2 are the same, $\text{id} : (X, d_1) \rightarrow (X, d_2)$ is trivially a homeomorphism, and thus for every such b , there exists $\delta > 0$ such that $B_\delta^{d_2}(b) \subseteq B_\varepsilon^{d_1}(a)$. Since $B_\varepsilon^{d_1}(A)$ is clearly equivalent to

$$B_\varepsilon^{d_1}(A) = \bigcup_{a \in A} B_\varepsilon^{d_1}(a),$$

it follows that for all $b \in B_\varepsilon^{d_1}(A)$, there exists $b \in B_\delta^{d_2}(b) \subseteq B_\varepsilon^{d_1}(A)$. Fix δ_b to be the sup over all such δ s, and we have that for all $b \in B$, there exists $a \in A$ such that $d_2(a, b) < \delta$.

I got stuck here. I also don't fully understand what it means for a set to be bounded in the question; my understanding after looking some stuff up is that no points in the set are infinitely far away from each other under a specific metric. But it's not clear how we define bounded in this problem, since there are two metrics. I didn't get to the point of the problem where boundedness is important so this didn't impact my proof, but I didn't quite understand the problem due to this.

Problem 7

Let $x \in X$. We have $\bar{d}(x, x) = \min \{0, 1\} = 0$. Moreover, if $\bar{d}(x, y) = 0$ for some $x, y \in X$, then $\min \{d(x, y), 1\} = 0$, and thus $d(x, y) = 0$, and thus $x = y$. Therefore, $\bar{d}(x, y) = 0$ if and only if $x = y$.

Let $x, y \in X$. Since d is a metric, $d(x, y) = d(y, x)$, and thus

$$\bar{d}(x, y) = \min \{d(x, y), 1\} = \min \{d(y, x), 1\} = \bar{d}(y, x).$$

Therefore, $\bar{d}(x, y) = \bar{d}(y, x)$.

Finally, let $x, y, z \in X$. We have

$$\bar{d}(x, y) + \bar{d}(y, z) = \min \{d(x, y), 1\} + \min \{d(y, z), 1\}.$$

We will brute force this with four cases.

- Suppose $d(x, y), d(y, z) < 1$. Then since $d(x, y) + d(y, z) \geq d(x, z)$, we have $d(x, z) < 1$, and thus

$$\bar{d}(x, z) = d(x, z), \quad \bar{d}(x, y) = d(x, y), \quad \bar{d}(y, z) = d(y, z).$$

Therefore, $\bar{d}(x, z) \leq \bar{d}(x, y) + \bar{d}(y, z)$.

- Suppose $d(x, y) \geq 1 > d(y, z)$. Then

$$d(x, y) + d(y, z) \geq 1 + d(y, z) > 1.$$

Since $\bar{d}(x, z) \leq 1$ for all $x, z \in X$, we have that

$$\bar{d}(x, y) + \bar{d}(y, z) \geq \bar{d}(x, z).$$

- Suppose $d(y, z) \geq 1 > d(x, y)$. By the same logic as previously, the result follows.
- Suppose $d(x, y), d(y, z) \geq 1$. Then $\bar{d}(x, y) + \bar{d}(y, z) = 2$. If $d(x, z) < 1$, then $\bar{d}(x, z) = d(x, z) < 1 < 2$. If not, then $\bar{d}(x, z) = 1 < 2$. Therefore,

$$\bar{d}(x, z) \leq \bar{d}(x, y) + \bar{d}(y, z).$$

Therefore, $\bar{d}$ is a metric on X .

Note that the basis $\mathcal{B}_{\bar{d}} \subseteq \mathcal{B}_d$, so the topology induced by $\bar{d}$ is a subset of the topology induced by d . It thus suffices to show $\mathcal{B}_d \subseteq \mathcal{T}_{\bar{d}}$. Let $B_r^d(x)$ be an open ball in the metric d . If $r \leq 1$ we are done immediately by definition of $\bar{d}$, so suppose without loss of generality that $r > 1$. Let $y \in B_r^d(x)$, and define $\ell := r - d(x, y)$. For any $z \in B_\ell^d(y)$, we have

$$d(x, z) \leq d(x, y) + d(y, z) < d(x, y) + r - d(x, y) = r.$$

Therefore, $z \in B_r^d(x)$, and $B_\ell^d(y) \subseteq B_r^d(x)$.

Using this same notation, note that $\bar{d}(x, y) \leq d(x, y)$ for all $x, y \in X$, since the only possibilities are that $d(x, y) = \bar{d}(x, y)$, or $\bar{d}(x, y) = 1$ when $d(x, y) \geq 1$. We thus have

$$B_{\ell_y}^{\bar{d}}(y) \subseteq B_{\ell_y}^d(y) \subseteq B_r^d(x)$$

for all $y \in B_r^d(x)$, where ℓ_y is the distance corresponding to y as defined above. Consider the union

$$U := \bigcup_{y \in B_r^d(x)} B_{\ell_y}^{\bar{d}}(y).$$

By logic we have used extensively in the past, since each $B_{\ell_y}^{\bar{d}}(y) \subseteq B_r^d(x)$, we have $U = B_r^d(x)$. Since this is a union of open sets, $B_r^d(x)$ is open in the topology induced by $\bar{d}$. Therefore, $\mathcal{T}_d \subseteq \mathcal{T}_{\bar{d}}$, and they must induce the same topology.